Essex Planning Leads

An automated data pipeline that monitors the Essex County Council Planning Portal, classifies construction planning applications by trade, and generates sales lead digests for subcontractors.

Features

- **Portal Scraping**: Periodically fetches recently received and determined applications, parses details and associated document metadata.
- **Rule Pre-filter**: Discards non-commercial applications (pure tree preservation orders, advertisement consents, routine environmental monitoring).
- **Trade Classification**: Tags relevant applications against a 10-trade taxonomy with confidence scoring, project scale estimation, and lead contact routing.
- **Digest Delivery**: Produces email-ready HTML and Markdown digests filtered by trade, district, and minimum confidence threshold.
- **Local Web Interface**: Lightweight dashboard for filtering applications and previewing digests in the browser.

Project Structure

```
None
essex_planning_leads/
├── config.py              # Central settings, taxonomy definitions, prompt
templates
├── database.py            # SQLite database access and markdown export
├── extractor.py           # Portal scraping and HTML parsing
├── prefilter.py           # Fast regex pre-filter for non-construction filings
├── classifier.py          # Trade classification engine (OpenAI + rule-based
fallback)
├── classify_all.py        # Batch classification runner
├── digest_generator.py    # HTML and Markdown digest builder
├── run.py                 # End-to-end pipeline runner
├── web_demo.py            # Local web dashboard
├── test_phase4.py         # Automated pipeline tests
├── requirements.txt       # Dependencies
└── data/
    ├── planning_leads.db  # Primary SQLite database
    ├── raw/               # Raw JSON records
    ├── markdown/          # Individual application Markdown files
    └── digests/           # Generated trade digest files
```

Setup

```shell
pip install -r requirements.txt
```

To enable live OpenAI classification, add your key to `.env`:

```
OPENAI_API_KEY=your_key_here
OPENAI_MODEL=gpt-4o-mini
```

If no key is provided, the system automatically runs with the local rule-based classifier.

Usage

1. Full Pipeline (End-to-End)

Run scraping, classification, and weekly digest generation with a single command:

```shell
# Standard run (scans latest 2 weeks)
python run.py

# Run on existing data without scraping
python run.py --skip-scrape --min-confidence 0.70

# Target specific trades
python run.py --trades Roofing,Scaffolding --limit 20
```

2. Generate Digests

Generate standalone digests for specific trades or areas:

```shell
# Generate sample digests for Roofing, Scaffolding, and Groundworks
python digest_generator.py --samples

# Custom digest for Chelmsford
python digest_generator.py --trade Roofing --area Chelmsford --granted-only
```

3. Web Dashboard

Launch the local web interface:

```shell
python web_demo.py
```

Open `http://localhost:8000` in a browser to search applications, filter by trade/status, and export digests.

Scheduled Execution

For automated weekly runs via Windows Task Scheduler or cron:

```shell
python run.py --weeks 1
```

Recommended schedule: Twice weekly (e.g. Monday and Thursday mornings).